

Active Inference BookStream #001.07 ~ "Governing Continuous Transformation"

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hello and welcome it's January 24th 2023 we're in active bookstream number 001.07 on governing continuous transformation off to you blue so we are the active inference Institute a participatory online Institute that is communicating learning and practicing applied active inference you can reach us at all of these social media links this is a recorded and archived live stream so please provide us with feedback so that we can improve upon our work all backgrounds and perspectives are welcome here and we will be following good video etiquette for live streams again if you'd like to participate act inference activeinference.org is our main website and there's a link to get involved and join many of the ongoing collaborative projects or live streams or both uh this is as Daniel said bookstream 1.07 so it's our seventh contextualization video um on this book governing continuous transformation by Bijan kesri so we will just jump right into it um I'm blue I'm an independent researcher in New Mexico and I will pass it off to Daniel all right I'm Daniel I'm a researcher in California and just happy to see this series continue tell and um to Tyler I'm a profile designer and our researcher in California great um so we are in part two free energy governance of this textbook um and the name of this chapter is cognition from superposition to reality um and so I think that that kind of hints to some Quantum stuff that might be elaborated more towards the end of this chapter and in the next chapter um and just a few keywords for this um chapter but still quite a bit of content here so Tyler do you want to read the start of the abstract do you mind sounds good can you guys hear me okay um all right attention base for you and inactivist approach address feg's cognition Dimension to this day strategy management research is not entirely each chain from its very path dependency of putting perception ahead of action I.E putting environment ahead of strategy while abv's structure cognition Nexus is fpg's critical building block ABV Iran erroneously trades action as a consequence of perception once we Embrace attention as action generated we start understanding that clever analysis and form of belief updating follows action generated stimuli not the other way around ABV is trapped in the old school

limitations opposed by bounded rationality in the discontinuous and distributed Quantum World agents boundary rationality is meaningless as enacted in for instance Powers the world to deliver supervisory signal effectively agents offload the burden of cognition to the environment boundary rationality is replaced by extended cognition the betterness of human cognition is effectively neutralized blue intake next one there are at least as many reality model models as there are economic actors every organization chooses akin to holding acting upon a light torch in the Darkness which environmental significance to attend to whatever the torch highlights is equivalent to breaking a Quantum wave of probabilities moving from superposition to Reality intermediated by the Markov blanket environmental enactment is essentially a purpose-directed interactive inferential and social learning sense making process all right so there are two sections to this chapter the first is attention-based View and the second is an activist approach um so Tyler do you want to take this slide sure all right so attention-based view so we actually touched on this a couple streams ago in the context of when we introduced this was that abhijan was going through different models of thinking about firm good cognition and they're all Limited in their own sense including attention-based medium we'll go deeper into why attention-based view is limited but also has important ways of thinking about how pharmacognition works and so the general idea of it is that attention is a limited resource and that permeates through all levels of the hierarchy of an organization so on the individual level you have limited attention that bows the the level the the balance or condition of what you're able to understand and perceive and decisions you're able to make and the boundedness of your individual cognition uh bubbles up into a social context which bubbles the kind of decisions a group of people can make and then that finally bubbles up into your organizational cognition which manifests through different rules positions players and resources um so this is just to say like hey we only have so many things that we can spend our time and energy on and that manifests at larger levels um in the kinds of rules that we make within an organization um blue Dano anything you would add here yeah so I can read this claim well attention-based views structure cognition Nexus is a critical building block of free energy governance fundamentally attention-based view suffers from severe shortcomings that weaken its relevance to practice by highlighting some of attention-based views most fundamental weaknesses we should further cement the case for free energy governance as a meta Theory connecting in circular causality structure cognition and capabilities once we understand how the Bayesian brain is building a model of its world we start seeing how anachronistic and full of friction are deep-rooted conceptions of organizational attention are

um do you guys have anything to add here okay cool all right so uh bounded rationality so this was um a concept that Bijan brought up here in this chapter and it was introduced um by Herbert Simon the term in like 1957 and I pulled this from the Stanford um Encyclopedia of philosophy so his um introduction of this term was a shorthand for his position against neoclassical economics and his call to replace the perfect rationality assumptions of homo economicus with the conception of rationality tailored to cognitively limited agents um and so like this little figure um here shows like the cognitive limitations and in in imperfect information and time constraints that we're faced with when we have to make a decision so we don't we always like are limited by what we can hold in our brain and like the view of the information that we have and like we were talking about like complexity um like Computing Pi like we could get a more accurate you know uh numerical digit of of Pi but but we're limited like by time like if we go on and on and on we could get more and more accurate but like why like is 3.14 good enough because I can pull it up in like a split second from my from memory um and so all of this like configures our bounded rationality and um we use a process that Herbert Simon introduced as satisficing to make a sub-optimal decision or like our best possible decision with the bounded rationality that we're faced with um and uh a quote from the um Herbert Simon Says broadly stated the task is to replace the global rationality of economic man with the kind of rational behavior that is compatible with the access to information and the computational capacities that are actually possessed by organisms including men in the kind of environments in which such organisms exist um Tyler do you want to read uh where this founded rationality concept is introduced in the chapter sure yeah so so I think that the context here is that he's trying to connect ABV to biographicality by saying all right well you have limited attention how does a firm decide where his attention is spent right like how do you make that decision collectively so he says in ABV top leadership's attention to the right issues and solutions consider the source the firm's competitive Advantage essentially is content rather than process center it confirms performance bottleneck remains upper at your mom's attention right on aggressionality the critical question then is how in the first place context is selected to determine the tension or have all them up Market signals connected with top down upper extra long expectations and predictions to determine attention and so what it stands like ABV doesn't really explain how that attention gets directed in the first place uh ABV is trapped in the old school limitations opposed about rationality and discontinuous and distributed Quantum World agents bounded rationalities meaningless as perceptual content it's given in inner predictions and this is precisely what allows the world to

over the supervisory signal effectively the firm is offloading the burden of condition to its environment so bound rationality is extended is replaced by extended cognition the Badness of human blue Daniel I think to add there yeah we'll go to the next slide where uh we'll display more about extended cognition so um Extended cognition I took this from this article by Mark spreeback uh extended cognition takes the idea that your mind is on your smartphone literally so it says that human cognitive States and processes sometimes spill outside of our heads into objects in our environment and this is something that we've looked at and talked about before in many ways like even with evil I'm not sure if you guys talked about it yesterday on the stream but in the thinking like a state live stream that we did we talked quite a bit about extended cognition and cumulative cultural Evolution and how like the the a lot of our cognition is like out there embedded in our environment so that we don't have to think about it um like when I grew up there was no internet so we would have to like you know I mean Colton like my son in the car the other day asked me some question and I was like you're gonna have to ask Alexa or Google and and he was like will you ask your phone for me and um it was like some fact question like how many bones are the body or something that I couldn't remember on top of my head and I said well wait till we get to school and then I'll ask my phone because my phone was like whatever playing music or doing and I'm driving right um and he was like that is so long like like you can't wait 15 minutes to get to school to like answer the question I was like man when I was a kid if I wanted to know the answer to this question like I would have to go to the library look in the card catalog find a book like or encyclopedia or something oh it's a goodness book of world record that's what it was it was like the world's smallest pancake or something like that um and so like we would have to find the book and then look it up in the book and it would take days like if we could even remember the question for that long and he was like oh my God where would you live so like this kind of extended cognition like we could just know everything with a click of a few buttons is amazing at extended cognition sorry I ramble all right um then maybe Tyler wants to talk about channeling attention cool so yeah I mean your extended cognition is not just uh you as this single known organization and it actually encompasses both your environment the The Firm around it and also maybe the environment around the firm like how's the attention of that cognition get channeled like how do you decide what matters what does it matter and so what John says is while ABV views organizations as systems of structurally distribute attention where stress for channel detention ABV trees the process of channeling as a black box so it's like a framework for thinking about how cognition is limited in organizational

context but it doesn't really tell you like what's going on for that reason it doesn't end up being super practical at least in the John's telephone so he goes on to say neither you EV nor you ABV both fundamentally top-down oriented Frameworks have given due consideration to the dynamic interplay between top management team TMT and the board of directors more specifically there's no transparent structural link between the upper echelon level and field intelligence for example the very process that governs the two layers interaction determining a firm's attention and actions remains opaque and so I think broadly what Beyond saying is that especially like the inactivist approach of thinking that how your actions and the environment determine your perception that's missing in ABV and that's a critical aspect of understanding how cognition happens in an organization but specifically the airplane between the board and the the management team I wish you previous streams he spent a lot of time describing why that's a really critical and important relationship uh for firm um blue Dale anything you you got on here so I would just add that like um or maybe emphasize the point that we talk about a lot about top management here and the board and that interaction and then like the interactions between different levels of the hierarchy like from the top all the way to the bottom and I think that that we emphasize the importance of like the the feeding forward and feeding back and like the interaction between the layers and how information is transmitted between them but there's no um real concrete um examples of how to to best do that um and I think that this is like like for me like not just in a firm but also like in the body like how is that information like transmitted like how does like my big toe prioritize like what goes to my brain like obviously if I kick something really hard and hurt my toe or like if I get like a hangnail or like a like ingrown toenail or something it really hurts but like otherwise my toe like I don't even know it's ever talking to my brain so it has to to like provide salience somehow um and how this is done and um I don't know that's so critical for me I think that that's like really at the heart of all of um the active and Friends hierarchical structuring stuff attentional coherence so um here bishan quotes um Ocasio and this is like a 2011 paper I guess or maybe book but he puts a lot of emphasis on Ocasio as like one of um like the people that like his work really rests strongly upon so I encourage you to read the chapter if you want to learn more or go straight to Ocasio himself so he says the interaction of top down and bottom-up attentional processes is also implicit in the wick and Sutcliffe concept of the quality of organizational attention and in the re-rope analysis of attentional triangulation combining attentional stability top down attentional vividness bottom up and attentional coherence combining both um and and he goes on to say that um attention-based you cannot explain attentional coherence and so getting at

what I just was saying about like the interaction between the top down and bottom up and like how that works um can be explained in a tension-based view maybe it can be explained in free energy governance or we can maybe get some guidance on how to best do that hopefully in the coming chapters we'll learn more do you guys have any comments hey I'm trying to read these claims for me Tyler sure so ABV Iran honestly treats action as a consequence of attention and perception according to ppf addition prediction processing framework actions determine in circular causality to stimulate that if your attention and sense making while ABV is limiting a critical foundation and looking cognition structure the question arises whether abv's attention will need to be redefined as something more granular and practical so tension not only is too diffuse it's a cognitive concept that also is hard to it's hard to meaningfully operationalize in the context of continuous strategic renewals so exactly what blue was saying on the previous slide it doesn't ABV explains like okay I understand why it's pension being limited uh shapes affirms cognition but like what do I even do with that information like how is that going to be impactful how can I model that um and the genre spending like well it's not that useful right now but he's teasing that maybe it could be with active inference or free energy governance thank you yeah like what is that something more granular and practical like how do we focus attention like how do we like break attention up into like sub components that are more granular and practical like um yeah like what is in our our field of view or what should be in our field of view like um what we're looking at what we're prioritizing and do those things line up I don't know um I could speculate all day but it's hard to to know really what that more granular and practical thing might be what one other quick note on that is there's the question of coherence and adaptivity for example that student answering their cell phone in the classroom on the previous cartoon that may have certain dimensions of adaptivity or fulfill certain preferences of One agent but not another and then there's this question of action orientation and so if we're always in the business of asking about the environment doing the market research not thinking about the strategy taking this sort of perception first mindset there still is the question of attentional coherence are we paying attention and sense making putting all the pieces together from what we're perceiving and also there is this inactive shift which is incomplete but it's part of the puzzle which is we want to be having attention regimes of attention on our action and that brings together the embodiments and the an active view with the attentional view which as brought up doesn't really describe what attention is and still can be mobilized in a passive setting but together we have this active attention and that flowers in active inference where attention is internal action so isn't it like pay attention to

what you're paying attention to is that like a nested little little hint there well in live stream 28 we saw a three layer model of attention and there is no single one way to model any cognitive phenomena it's going to be up to people in different settings to understand what treatments of attention are insightful or useful or applicable I don't know I think that that like paying attention to what you're paying attention to um kind of gets like a lot of attention in like the business context or like even in like um prioritizing like your to-do lists right like are you just like fighting fires all the time or are you like making progress to do like things going forward making progress on your own goals so I think like I don't know that might be might be a critical um like governing variable right that um so let me read this claim in that regard predictive processing framework in in the form of inactive inference is potentially a critical complement to and possibly substitute for the attention-based view predictive processing framework provides the concepts and Logics that give new meaning to the linkage of structure and cognition according to predictive processing framework prediction models and predictions self-optimize as a matter of pursuing Surprise minimization by way of free energy minimization it is strictly a priori not an attention challenge but a purpose-directed prediction challenge the evolution of self-driving Vehicles provides an exemplary illustration of this point once the challenge was conceived as a prediction rather than attention challenge the road to Perfection is simply a matter of continuous and purpose-directed prediction error minimization and I don't know like what is really meant by this like like the challenge of self-driving was conceived as a prediction like once people predicted that it would happen um I I'm not really sure like what Bhajan means here I could maybe take a stab at that one um because I think what he's saying is that like for a self-driving car that's kind of thing that if you were being using purely linear thinking only using the things you know already about the world like you would never a self-driving car wouldn't be like an obvious Next Step but you have to take this leap of faith in action-oriented week of faith into saying no we're going to build a self-driving car and then once you've decided that that's your your path you're gonna go down now it's just like optimization on that path and so I think this he was after like illustrating the inactivist approach of like hey a self-driving vehicle is nothing like obvious or self-evident about that needing to happen but once you decide it's going to happen then it's a prediction problem agree there's different readings one way also not knowing the self-driving car game if we were in this passive perception first world then there would be uh the main issue to resolve would be understand from noisy sensor readings where is the location of a person and producer uncertainty about where the person is or what that object is however once we see it in terms of

this continuous purpose-directed prediction error minimization we're going to Center the role of action gas break and turning and a few other affordances that cars have so instead of going for all the decimal points on whether that's a person or a mailbox or a cone what does it look like to take this unfolding action-oriented view where you're aligning with your preferences for example Not to cause harm to people and then lesser priority Not to cause harm to objects maybe steering away from that object you can have a good policy posterior on it's a good and likely action without having all the decimals of precision about the environments so once we understand how essential action is to the self-driving car it reframes what we have to do with the sensor data and it centralizes the relevance of the strategy yeah I do remember something about like um some change in self-driving like in the sensor system so maybe that like references the um like attention versus prediction problem or maybe that's like being referred to internally in there I'm not sure okay so that sums up the attention based view section and we're moving in now to um the inactivist approach um so this is like at the start of the chapter um quotes Thompson 2007 and says the organism's environment is the sense it makes of the world the environment is a place of significance and Valence as a result of the global action of the organism I'm so again taking this like action-centric perspective um and he quotes ramstead of Dallas 2020 by saying active inference is an active inference and he says indeed the essential question is how the firm brings forth significance and Valence that does not pre-exist but is the result of action-centric and socially constructed engagement with the outer World equivalent to internally assigning probabilities to the external hidden unknown states in the form of a Marco blanket this revealing of significance and Valence then lays the foundation for generating shared meaning and shared expectations in the form of unique interpretation models belief systems underlying firm performance and survival oh yeah so we can go on to see what else he says so he makes the claim dafton week 1984 differentiate between environment driven and interpretation-driven approaches to strategy making um and in this big quote they say in environment driven frames top managers experience environments as Concrete Hard measurable and determinant on the other hand top managers of firms confronting more unstable or unanalyzable environments are thought to construct course or enact a reasonable interpretation that makes previous action sensible um thus they attempt to construct their environments through their strategies rather than developing strategies in response to environments resulting in strategy environment beliefs or proactive Logics and he also says despite wic's strong influence on Strategic Management research to this day the environment-driven approach remains dominant um so I think that this is like one

of the central themes that's emerging in the book is um that that like the environment is somewhere out there it's some very real thing uh it's taken to be like it's it is this way it's always going to be this way um and it's like um and kind of something that Daniel and I have talked about at length like the niche as responding to the organism the niche response to the organism just as much as the organism responds to the niche and whether or not the niche includes other organisms in the environment or not is kind of irrelevant when we have things like books and smartphones uh so this environment is is changing um and it's changing maybe so fast and so much that this environment driven framework is perhaps not effective so um unstable or unanalyzable environments is maybe like this discontinuous environment that Bichon speaks of um because the the environment's changing fast like maybe faster than it's ever changed before because of like our Global connectivity and many other reasons so so now maybe all environments are unstable or unanalyzable or or more unpredictable than they've been before so that this um constant update like or attempt to attempting to construct the niche rather than viewing it as something stable is maybe more important for firm persistence glad you guys have any comments here yeah I mean I just think on the the second half of that of like very unstable and unanalyzable environments that really resonates in the start of that context where uh a lot of times our families will have no pretense of knowing anything that's going on in the environment like they're very uh they're very clear about their ignorance of the world instead they run small experiments to kind of gauge what's going on in the the external world and then kind of construct the world view from there rather than saying hey we have this great business idea let's scale this thing it's more like hey let's discover who our users are um which I think really uh corresponds well with that second half that were also with our classical temperature example let's just say that we were in a house and there was a heating element if we think that over the time scale of days things aren't going to change that much we might look towards a strategy of turning on and off the heater so dealing with the environmental variability in the niche we're in whereas if we're in an open space so there's no point in having any strategy of trying to heat the local Air because it's just going to get blown away in the next second then you might Center decision making around like finding a warmer location and so when there's a more stable environment then it puts us into a mode of like understanding the variability patterns the Circadian patterns and temperature and responding with what we have or maybe inventing something new to respond to those regularities but when there's the discontinuous and complex environmental settings that this work is focused on It Centers this kind of creating one's own Niche through action not just modifying it more

locally to do the best one can oh um Tyler do you want to read this claim do you mind sure so every organization chooses can holding a white torch in the darkness which environmental significance to act upon and tend to whatever the torch highlights is equivalent to breaking a Quantum wave of probabilities moving from superposition to Reality intermediated by the Markov blanket in this regard strategy research is challenged to be less mechanical mechanical mechanistically determined environmental strategy but more Quantum strategy to environment the quantum world is more tenuous than the one I imagine and this is actually quote from rebelli and this is not Sean uh the quantum world is more tenuous than the one imagined by old physics it is made up of happenings get discontinuous events without permanence is a world of fine texture intricate and fragile Venetian lace every interaction is an event and it is these light and ephemeral bends that weave reality not the heavy objects charged with absolute properties that are philosophy positive and support of these events oh thanks um so uh bishan goes on to say and I think we'll we'll come back to this um Quantum because I have some questions here so we'll come back to this at the closer to the end of this section um but right now let's read this claim um Bijan says the critical consideration is how an organism or organism sorry too much biology how an organization transmorphs the sensing and sense making activity at the individual level into a firm's institutionalized environmental enactment as a basis for continuous strategic renewal X-Men is essentially an interactive and generative social learning sense making process hence inactive inference done at AI elegantly addressed this micro macrodichotomy they say while hypothesis setting is an individual activity hypothesis testing is an organizational activity according to smurchic and stubber organization members actively form enact their environments through social interaction a pattern of enactment establishes the foundation of organizational reality and in turn has effects in shaping future enactments the task of strategic management in this view is organization making to create and maintain systems of shared meaning that facilitate organized organized um do you guys have any comments here one thought I'm imagining uh a fleet of boats the upper echelon View and the board of directors specifically they may not need to be concerned with the minutia of move the wheel left move the wheel right on this boat at that time but they are as it says here the task of strategic management in this view is organization making so that justifies some of this upper echelon Focus it's not coming I don't think from a place of elitism or chauvinism about those upper management groups but it's especially targeting to those who are in a position to be involved in organization making which isn't steering the boat but allowing the agency and enactment to happen in the front line requires those not in the front

line to take The View that is being promoted here rather than just imagine if we had enough sensors on enough boats we could have all of them viewed up in front of us and then we could call out who should turn left and who should turn right with the three to eight board meetings per year it's not going to happen so what does the organization making that responds to that look like I'm kind of curious about what you just said because something I've been wondering about through this whole book is what our board is really necessary for that meaning making or whether that can be done in a more distributed way and I think the Jean tests on this a little bit in our last call with him saying that like well this is their job is to be the board's job specifically should be able but distinct from the operation so like they're in this unique position to really build meaning and create sense making a facility that process but I'm wondering if it's no like actually necessary for that to happen because it does strike me as like very top down like imposing a sense making framework on organization is a ton of power uh and it does very much constrain the cognitive abilities of the larger organization and it strikes me just on this simple reading of it that like maybe something is missing from that or maybe a potential other way forward that is less top-down as possible but I'm probably misunderstanding Bajan as well so I'm curious what y'all think it'll be so fun to explore but let's just say that the task of strategic management is organization making maybe this is a saddle point View that everybody can get on board with but then we go great upper Management's role and function was organization making and now everyone has access to the tools and at least view or comment access to the process of organization making so transforming the understanding of what the upper echelon does in terms of organization making so just double checking if you're not involved in steering the ship you're involved in organization making and action orientation and then perhaps that function could be taken on by another organizational structure with that as its focus or that could be an emergent functionality of other organizational units but it's really hard to go from Vision coming from the top to Simply a vision that's coherent arising from the bottom and from the outside and so there's an intermediate view that qualifies the conversation differently yeah I think that makes sense so I think we have observed that since making happening both in the bottom and the top and especially in tech companies um where like you will have a board and they are making bit like you know top-down decisions but you will also have like Consultants almost embedded at each level of the hierarchy who do that sense making every level of the hierarchy uh and bubble that up also to the top so I think it does exist where you kind of have it's not purely sense making happening at the board of their TMT level so I just want to point out the first

sentence here and it really resonates with what you just said entirely like the way that I've been reading this he says the critical consideration is how an organization transmorphs the sensing and sense making activity at the individual level into a firm's institutionalized environmental strategic renewal like what I took that like individual level as a very like bottom-up thing um because it's like you know each like little worker Aunt right in the colony like I I took that as like a very bottom-up and distributed thing like how do you take that and turn it into like some kind of collective sense making but really that could also be like there's an individual like level at the top like each board of directors is selected for their different expertise right so like you have a variety I mean ideally you have a variety of different views and perspectives at the board level and then how do those individuals like translate their sense making into like something that the institution can then act upon um so I think that that is a very like it rings very true that like this individual um sensing and sin speaking is happening at like the top end at the bottom um I think that that's that that brings true though I just want to draw one distinction based on some experience of like doing the top down since making so like I was a strategy consultant for a while and a lot of times you would do the top-down uh sense making where you would like talk to all levels of the hierarchy and you kind of act as the board in that situation and kind of Bubble Up all of this information into like a top level um idea about what's happening in the organization what the narrative is and what actions they should take and so it's like very top-down structured but there's like other models of doing this where each node and organization is more or less operating independently and since making it dependently whereas like the top-down structure isn't as important and maybe you're saying the same thing but it strikes me as like a little bit different where sense making isn't happening through a bubble up process through a pyramid but rather happening in like a web okay in a mesh and that strikes me as like an important difference but maybe John is saying the same thing I'm not sure it's hard to know and it's hard to also know like how um you know when you have like even at like a single layer of the hierarchy like you have you know 25 people and like you know 12 of them think one thing and 13 of them think another thing like how do you like make sense of that like where there's this like opposing Viewpoint how do you collectively like get buy-in to proceed with like either strategy a or strategy B like even at a single level um and then pass that information up or down without like well I wanted it to be this way or I wanted like the people going um that like the opposing view like whatever wins right like in a vote the 13 would win if that's the way that that it's run or maybe there's weighted voting or who knows right like however a firm would structure itself but it's like

really important I think to get strategy buy-in like from everyone at least like hear all of the points in order to to proceed and and have like the Viewpoint adopted widely and like get people to get behind it so it's I don't know it's a hard like weird thing to think about how that information is passed up or down in the hierarchy Tyler do you have something out well I was just going to add that this is a really important point like I don't think I'm not just trying to be like pedantic about it like it actually a lot of organizational strategy debates hinge on this of whether really you can do sense making in a like hierarchical way like truly like a pyramid or like a more mesh Network so like the mesh networks are more like teal organizations or democracies and like this is a very contentious point and I was hoping that we could like figure it out head on or understand the Jean's like perspective on it and I think he thinks both are possible because I think he's hinted at that in the previous live streams but he hasn't said it like uh super clearly so far um but something I've been like very eagerly waiting for him to like spell out clearly so maybe he'll get to it in the next chapters so Tyler like I have a request like in the next live stream you have to put in like a slide down hypocrisy and like explain that to us like in a very like break it all down break down all these different like strategies for us because like you have that expertise and I'm lost so that there's there's my special request okay awesome yeah I mean I've just been in like lots of debates with even within especially in Dallas where they try to do the mesh Network version of things not the the pyramid and a lot of times just like a total disaster and it doesn't work but there's always this dream especially in Dallas like oh could this be possible like do you actually need the hierarchy at all can this work more like a spider web rather than the pyramid um and I've been eating this whole book I've been like waiting for the answer to that so I'm excited to ask for joins yeah and and also just getting these dimensions sketched because what if it's like well yeah we can have a slime mold but then we may split a lot and not all the things that split off may survive if that is tolerable or preferable to you slime mold it is but if we all wanna ride or die together then this kind of a structure is what we want so making sure that there's like consent and transparency around what people are wanting to be in because also in sort of emergent online 2020s eras teams and Dows it's like are we hanging out in the group chat are we building a product that's going to change the information environment are we learning about other projects are we learning about other early projects invest in like it can move very rapidly amongst those spaces and at the risk of trying to capture for birds and getting none there's all these other failure modes but maybe again that's what those people wanted to do in that blanket at that time but when you have shareholders in a board and there are some responsibilities that come

into play and these sort of okay to fail or okay to Sunset experiments in online organization hopefully can reveal empirical patterns that we use to carry forward organizational technique yeah my reads so far and what John's doing is that he's saying like yes sure a slide mold organization is a totally thing you there's a thing you could conceive of and like that's a valid way for thinking about an organization but he's specifically thinking about like the legal context that a lot of organizations live in today and making uh active inference or free energy governance work for those organizations in the world that we live in so I'm not sure if that resonates with you guys but oh okay nice nice uh debate tangential I don't know information dive all right uh um Bijan makes this point and he says organizations are challenged to better distinguish how their actions differ from those of others as they construct econishas and he um really kind of gives this like um like a lot of back a lot of like background information but I just pulled these two quotes about like distinguishing self from other like how do you distinguish what is me and what is you and also like about how do you sense make like help others see what's in your head um and in in talking about these things he gives two examples which I thought were really cool from biology so I wanted to read them he quotes Godfrey Smith 2016 and says whenever a fish emits a pulse it also sends a copy of the command around to the sensing system enabling that system to counteract the effect of the pulse it has produced The Fig the fish is tracking and registering the distinction between self and other between the effects on its senses of its own actions and the effects of events going on around it um and then he also goes on to say to use the example of sharks he says sharks pequotes lonza at out this 20 20 paper he says sharks have organs called lorenzini blisters which sense electrical Fields dolphins and tooth whales for example benefit from echolocation abilities whose sound pulses can penetrate soft tissue to provide them with an x-ray like mental image of the objects of interest in order to tell the other Dolphins what they have discovered they actually create a visual picture of what they just saw in the minds of other Dolphins so I didn't like put a full like explanation of echolocation because it gets complicated but like here's this dolphin and it sends out these echolocation rays and if something is in its echolocation site because it's not really sight right but if something it sees it then the arrays that are reflected back like give like a are like molded around so you can imagine like Whipple ripples through like water right like so they mold around like a rock or a stream and so what comes back back is actually like a physical like Ripple picture of the rock or whatever it's seeing and so it can actually then modify its own like echolocation sounds to give a picture of the rock to someone else which I think is so cool right like like to a dolphin friend no not like someone else um

because I couldn't see a ripple picture of a rock but maybe it's dolphin friends could um anyway I think that that is a neat idea of how to like really um you know we always ask people like do you understand me do you know what I mean do you really know um but like it's really hard to know perceptually what someone else is is what's behind their markup blanket right sorry I'm just the one fascinated with biological sensing and Sun speaking okay uh Tyler do you want to read this claim for us sure so more specifically optimization is essential across the consist constituent elements determining the circular causality of the underlying and active inference one actions generating reliable sensory input Precision two the ability to express beliefs that bad beliefs consequences sophistication to enact only a bounded set of external States three leveraging salience without missing the very error signaling value that non-selling stimuli May nonetheless behold and four foresight to anticipatory develop and strengthen the internal States I.E firm resources in form of capabilities critical to Resource recombination so sensing sense making and Duality management cool I I find it interesting here that one two and three are within the life cycle of an organization or an organism they refer to the reliability of action the ability to have an appropriately expanded vocabulary in terms of sophistication to describe relevant action and then three essentially attention which is to make sure that you're not just falling victim to hyper skewed or hypernormal stimuli but rather you're using your attention to enrich incoming stimuli and also not ignoring what you're not totally paying attention to that might be on the margins of relevance but then four is framed here as an anticipatory question in the long run whether the capabilities are successful to manage resource for combination that's the question of life and death so one two and three are like the behavioral imperatives and then four is kind of like the organism looking to the Horizon maybe it makes it or maybe its offspring does or maybe none of the above um and then just this final um quote from from the the paper or from the book he says Duality is an essential quantum concept that underpins the Mastery of managing the distributed and discontinuous nature of today's market realities um and like I'm curious as to like what Bhajan is referring to as Duality like in the quantum context I always think about like wave particle duality but I think um he has talked here a lot about superposition and maybe he's referring to superposition um maybe he means superposition like you can exist in two different states you know until until for a long time like you can be both black and white one and zero um so I'm not I'm not really sure what he's referring to as Duality but I think maybe we'll get into it in the next yeah I was curious whether the connection he's making here is like one of analogy saying like hey in the same way you don't know what what's out there in the quantum world do you only You

observe something and you kind of collapse all of these waves into electron is it an analogy in the active inference free energy governance world with through you know the inactivist approach or is he saying that this is actually like the same thing that is the same principle underlying and I wasn't totally sure which way what he meant there yes we should explore there's the wave particle duality famous there's the Observer observed Duality or the experimenter environment duality there's multiple dualities and then if we think back to Bayesian physics what does this help us with let's go into it more in the next chapter and then see how it actually connects back otherwise I believe it puts us squarely into the Quantum business advice genre but I know there's more to it than that would be like a very uh sellable like Quantum business advice Quantum business strategy manager um that's my new job I'd be like it's a heck of a consultant title am I working am I not working like you could check you still might not find out am I here am I there where am I actually um you can't measure my position and my momentum at the same time um that's too funny okay all right um I think uh we've reached the end of our set of slides I'm looking forward to exploring the quantum in more detail um as I find the quantum Concepts fascinating I guess bishan also does do you guys have anything else to add Before We Say Goodbye foreign next week thanks y'all farewell